

premium buyers must provide or discount sellers grant to complete the order within a specified timeframe.

In economics, I-Star represents the incremental cost incurred by demanders resulting from a supply-demand imbalance. We depicted this above via a graphical illustration. Following that example, our I-Star cost is determined directly from the imbalance Δq and the corresponding change in price Δp , that is, $I^* = \Delta q \cdot \Delta p = \Delta q \cdot (p_2 - p^*)$ (Figure 4.2d).

The variables of the instantaneous impact equation are:

Q = market imbalance (the difference between buying and selling pressure)
 ADV = 30 day average daily volume (computed during exchange hours)
 σ = 30 day volatility (day-to-day price change)
 a_1, a_2, a_3 = model parameters (via non-linear regression analysis)

Market Impact Equation

$$MI_{bp} = \underbrace{b_1 \cdot I^* \cdot POV^{a_4}}_{\text{Temporary Impact}} + \underbrace{(1 - b_1) \cdot I^*}_{\text{Permanent Impact}} \quad (4.7)$$

Market impact represents the cost that is expected to be borne by the trader based upon the underlying execution strategy, e.g., percentage of volume (POV), trade schedule, etc.

The variables of the model are:

I^* = instantaneous impact
 POV = percentage of volume trading rate
 b_1 = temporary impact parameters (via non-linear regression analysis)
 a_4 = model parameters (via non-linear regression analysis)

Market impact further consists of the temporary and permanent cost component.

Derivation of the Model

Consider a situation where buyers have V shares to buy and sellers have V shares to sell—both within the same time period and urgency needs. In this situation we have an equilibrium condition where the shares to buy are equal to the shares to sell. Therefore, we expect there to be V shares transacted in the market without any extraordinary price movement (but there may be some price movement due to market, natural alpha, or noise).